

Section: TuTh 2:15PM - 3:30PM, College of Business 127

Prerequisites: AE 201, ES 305

Instructor: Roohi Golkhatmi, Ehsan roohie@erau.edu, LB 227

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[Links to an external site.](#)

(please start the subject line of your email with AE 319 for all e-mails)

Office Hours: TuTh 11:00-12:00, 3:30-4:30, MoWeFr 3:00-5:00.

TAs: To be announced.

Help session To be announced.

COURSE DESCRIPTION:

Conservation equations and fundamental fluid dynamic principles, elementary solutions of inviscid incompressible flows, Fourier series, flows around airfoils and wings, introduction to viscous flow and turbulence. Review of thermodynamics to understand compressibility, normal shock waves, oblique shock waves and expansion waves, compressible flow through nozzles, subsonic and supersonic flow around airfoils, including linear theories, elements of hypersonic flow.

COURSE GOALS:

This course introduces aerodynamics to Astronautics and Rocket Propulsion track students. Emphasis on understanding fundamental concepts both on low- and high-speed regimes.

LEARNING OUTCOMES

1. Apply global conservation of mass and momentum to engineering systems
2. Apply Bernoulli's equation (relating pressure and velocity)
3. Compute the lift, pitching moment, and pressure distribution for an incompressible airfoil using thin airfoil theory and panel methods
4. Understand the importance of viscous effects on the flow around an incompressible airfoil
5. Calculate induced drag as well as lift distribution on finite-aspect ratio wings using lifting-line theory and the vortex lattice method
6. Analyze basic supersonic flow fields using the normal shock, oblique shock and Prandtl-Meyer expansion relations.
7. Evaluate flow field properties in compressible flow through nozzles.

8. Scale incompressible airfoil data to subsonic Mach numbers and estimate the critical Mach number.
9. Use both exact and approximate theories to calculate surface pressure, lift, and drag for supersonic airfoils.
10. Calculate surface pressure, lift, and drag on a hypersonic body using approximate theories (Newtonian and modified Newtonian).

Required Textbook:

Text/Tool: Aerodynamics for Engineers
ISBN: 9781009098625
Authors: John J. Bertin, Russell M. Cummings
Publisher: Cambridge University Press
Publication Date: 2021-08-12

Reference Materials

Text/Tool: Multimedia Fluid Mechanics DVD-ROM
ISBN: 9780521721691
Authors: G. M. Homsy
Publisher: Cambridge University Press
Publication Date: 2008-04-28
Link: <https://www.cambridge.org/core/books/multimedia-fluidmechanics-online/643A1961C05C52488B764C82E0EA76EA>

Notes

If you are on a university IP you can access this book for free.
From off campus first search for the book in the hunt library search engine. Once you find it you can access it through the link in the search engine by logging in.

Text/Tool: Foundations of Aerodynamics
Authors: Arnold M. Kuethe, Chuen-Yen Chow
Publication Date: 1986-01-01

Text/Tool: Aerodynamics for Engineering Students
ISBN: 9780081001943
Authors: E. L. Houghton, P. W. Carpenter, Daniel Valentine, Steven Collicott
Publisher: Butterworth-Heinemann
Publication Date: 2016-08-26

Text/Tool: An Album of Fluid Motion
Authors: Milton Van Dyke
Publisher: Parabolic Press, Incorporated
Publication Date: 1982-01-01
Link: https://web.mit.edu/swangroup/footer/album_fluid.pdf

[Links to an external site.](#)

Text/Tool: Introduction to Aerospace Flight Vehicles
ISBN: 979-8-9852614-0-0
Authors: J. Gordon Leishman
Publisher: Pressbooks
Link: <https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles>

[Links to an external site.](#)

Text/Tool: Visualized Flow
ISBN: 9780080340654
Authors: Yasuki Nakayama, W. A. Woods, D. G. Clark, Japan
Society of Mechanical Engineers
Publisher: Pergamon
Publication Date: 1988-01-01

Text/Tool: Fundamentals of Aerodynamics
ISBN: 9781259129919
Authors: John D. Anderson, Jr.
Publisher: McGraw-Hill Education
Publication Date: 2016-03-21
Assessment Activities

COURSE STRUCTURE

You should review/study the material before the lectures. A PDF file of lectures (incomplete) is provided on Canvas.

In the classroom I will be also be doing example problems relevant to your current weekly homework assignment, or an upcoming exam. The solutions to the examples that I go over will be posted to Canvas as well.

Exams:

There will be two exams. The first Exam will cover incompressible flow and the second Exam (given at the day of the final exam) will cover compressible flow. There will be no make-up exams except under extraordinary circumstances and with the instructor's approval.

All exams are open text only (no notes).

Exam 1: 10/10 1:00-3:30 PM Please note the **early start time: 1:00**

Exam 2: Day of the final exam

You will be given an extra 5 minutes to scan and post your exam (in PDF format) to the corresponding Canvas assignment.

Guidelines for Homework:

Your work must be neat and easy to follow and in PDF format (no paper submissions!). If it is not, no credit will be given. Scan homework in PDF format (do not send photos) and upload to Assignment on Canvas.

1. State briefly and concisely (in your own words --- do not just repeat the problem statement) the information given and draw a schematic, if appropriate.
2. State the information to be found.
3. Give the appropriate mathematical formulation of the basic laws that you consider necessary to solve the problem.
4. List the simplifying assumptions that you feel are appropriate in the problem. (Sometimes it is not clear ahead of time what assumptions are needed to make a problem tractable. In such cases, proceed to solve the problem and take note of the assumptions you make along the way.)
5. Carry the analysis to completion algebraically before substituting numerical values.
6. Substitute numerical values (using a consistent set of units) to obtain a numerical answer. The number of significant digits in the answer should be consistent with the given data. (All numerical answers must have the appropriate units given, unless the answer is nondimensional.)
7. Check the answer and review the assumptions made in the solution to make sure they are reasonable.
8. Label (box) the answer. Many times the answer will be a formula rather than a number. You should check that the units are consistent in each of the terms, and that reasonable answers are obtained in limiting cases (by letting variables take on large or small values).

Homework will be due on Fridays at midnight. The last 2 homeworks will be due Wednesdays. An important task is reviewing the homework solutions posted on canvas

Grading:

Exam #1 40%

Exam #2 40%

Homework 20%

Letter Grade Percentage

A 90 - 100%

B 80 - 89%

C 70 - 79%

D 60 - 69%

F < 60%

UNIVERSITY POLICIES

Academic Integrity

Embry-Riddle is committed to maintaining and upholding academic integrity. Academic integrity violations include cheating, fraud, plagiarism, and double-submissions. More specific definitions of these violations and their consequences are described in the Dean of Students' Honor Codes and Student Policies. To ensure fair and full achievement of degree requirements, students must prepare and present their own work. To show that they have completed their work with academic integrity, students should keep any drafts, notes, calculations, and the like.

Use of Artificial Intelligence

The use of ChatGPT and other artificial intelligence (AI) tools in an educational setting must be guided by ethical principles and academic integrity. Students and faculty may use these tools for research support or aids to enhance their learning and scholarship but should not rely solely on them to produce original work. The University recognizes the potential benefits of AI but acknowledges the potential risks, such as creating unauthentic, inaccurate, biased, or harmful content. Students should follow best practices to ensure authentic scholastic accomplishment and academic integrity, including avoiding plagiarism or machine ghostwriting. Ultimately, the University upholds the importance of intellectual honesty and ethical research practices and expects students to act in accordance with these principles when using AI tools.

Course Policies

The University Honor Code will be maintained. You are responsible for knowing and adhering to the Honor Code as well as campus, university, and department policies and guidelines. You are encouraged to discuss assignments with your instructor, the teaching assistant (if any), and your classmates. Unless stated otherwise, all work submitted for credit must reflect your own work and understanding of the course material.

Electronic Devices -- All electronic devices must be placed in airplane mode (if used for note-taking) or out of sight during lectures. Student(s) will be asked to leave the remainder of the lecture if these devices disturb the lecture. Electronic devices shall not be used during quizzes/tests/exams. If an electronic device is not in your bag during these you will automatically receive a zero

Problem sets/homework

Problem sets (homework) are due at the time indicated on the canvas.

All home submissions will be online as a single pdf file. No late homework will be accepted. If a legitimate need arises late homework may be allowed with prior written (email) permission.

Solution sets for the homework will be posted on the Canvas website.

No working on homework during lectures.

Collaboration on homework is allowed. Collaboration does not mean emailing computer codes to each other, lending/borrowing homework, copying old homework solutions or someone else's homework, etc. It means offering someone help when they are stuck on a problem or explaining a concept to someone so that they can proceed to solve the problem or code a program on their own. If you have any doubt if a certain act constitutes collaboration please clarify with the instructor. The work that you turn in must be your own work. In case the submitted work is identical (or shows other evidence of lack of integrity), zero credit will be given and the students will be reported for dishonesty.

There are tutor services available online. These tutor services have frequently provided incorrect answers when students have used them in the past. Any blind copying of tutor-provided answers will be treated as an act of plagiarism and will be referred to the relevant responsible academic integrity authority.

All solutions, notes, codes, etc. provided are for personal educational purposes only. Posting or sharing of such on any forum may be a violation of the Digital Millennium Copyright Act.

Tentative Schedule AE 319 Aerodynamics – Fall 2024:

Week	Topic	Reading	Notes
1- 08/27 – 08/29	IM0: Introduction IM1: Conservation laws	2.1 – 2.5	
2- 09/03 – 09/5	IM1: Conservation laws IM2: Potential Flow	2.7 – 2.10, 3.1 – 3.6, 3.8 – 3.10	HW #1
3- 09/10 – 09/12	IM2: Potential Flow	3.11 – 3.16	HW #2

3- 09/17 – 09/19	IM3: Viscous Flow	4.1 – 4.2, 3.13 -3.14	HW #3
5- 09/24 – 09/26	IM4: Flow around Airfoils	5.1-5.2, 5.4, 6.1-6.5	HW #4
6- 10/01 – 10/03	IM5: Wings	7.1-7.3	HW #5
7- 10/08 – 10/10	Review		Exam1
8- 10/15 – 10/17	CM0	Notes	
9- 10/22 – 10/24	CM1 Introductory Concepts	8.1 - 8.2	HW #6
10- 10/29 – 11/31	CM2 Shock Waves	8.6 – 8.10	HW #7
11- 11/05 – 11/7	CM3 Nozzles	8.3 - 8.4	HW #8
12- 11/12 – 11/14	CM4 Subsonic and Transonic Flow	9.1 – 9.5	HW #9
13- 11/19 – 11/21	CM5: Supersonic Airfoils	8.5, 10.1, 10.3	
14- 11/26 – 11/26	CM6: Hypersonics [short week]	12.1-12.3	HW #10
15- 12/03 – 12/05	CM6: Hypersonics Review	12.4 – 12.6	HW # 11
16- Exam Week	Exam 2 (day of the final exam)		

Note: Sometimes modules will start the last lecture of the previous week. A weekly announcement will summarize weekly readings and assignments. Homework sets are due Fridays except for the last two sets which are due Wednesdays.